Local Income Taxes and Residential Sorting
Evidence from LEHD Origin–Destination Data

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April 18, 2026

Motivation

- Tiebout (1956): households "vote with their feet" across local jurisdictions offering different tax/service bundles.
- Strong evidence at the state level (**??**) and for sales tax cross-border shopping (**?**).
- But little causal evidence on local income tax sorting within metros — where moving costs are small and boundaries are frequent.
- Two data problems: local tax jurisdictions are sub-county (IRS/ACS too coarse), and credit systems + enforcement complicate effective tax burden.

This Paper

- Question: Do local income tax rate changes cause residential sorting across municipal boundaries?
- Data: LEHD LODES origin–destination, linking residence tract $\leftrightarrow$ workplace tract annually, 2002–2019, for OH and PA.
- Outcome: `share_live_in` = (workers who live and work in city i) / (total workers in i).
- Design:
  - OH: clean extensive margin (taxed cities vs. untaxed townships).
  - PA: universal EIT with rate variation + Act 32 (2012) enforcement consolidation as natural experiment.

Institutional Setting

- Ohio: municipalities may levy 0.5–3.0% on earned income; townships cannot. 302 increases, 45 decreases in our panel.
- Pennsylvania: universal EIT (Act 511, 1965); typical 0.5% muni + 0.5% school district. 291 increases, 162 decreases.
- Credit system (PA): worker's liability is $\max(\tau^{\text{res}}, \tau^{\text{work}})$. Moving without changing jobs yields no tax savings.
- Act 32 of 2008 (eff. 2012): consolidated collection in 67 Tax Collection Districts, mandated employer withholding. Before 2012 the tax was largely unenforced on nonresidents.

Preview of Findings

- Rate increases reduce `share_live_in` by 0.5–0.8 share-points per 1 pp rate increase, building over 7–8 years.
- Response is asymmetric: increases push out, decreases do not bring back.
- Effect is entirely post-Act 32: precise zero before 2012, large and significant after.
- City-wide relocation, not boundary arbitrage.
- School district EIT has the opposite sign: higher SD rates attract residents.
- Fiscally productive: EIT revenue +67%, total tax +30%, population/property values unaffected.

Data Sources

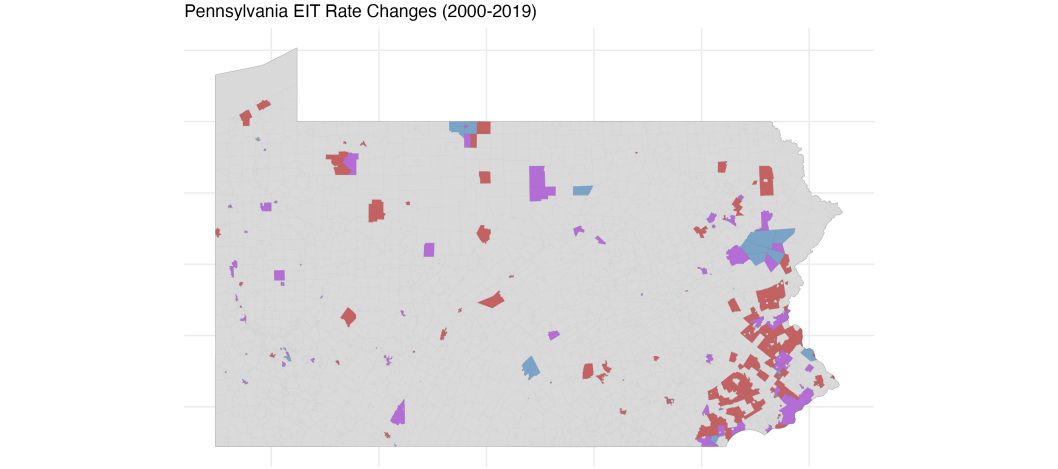
- LODES v8 (2002–2019, trimmed pre-COVID): tract-pair-year OD flows, main + aux files, segments SE01/SE03/SA01-03/JT04.
- OH rates: "The Finder" CSV, direct PLACEFP join.
- PA rates: scraped DCED MunStats (resident/nonresident muni EIT, school district EIT) ∼2,500 munis/yr; name-based COUSUBFP crosswalk (99.6% match).
- PA fiscal controls: DCED Annual Financial Reports (71 cols: EIT rev, police, streets, debt, state aid, pop, MV).
- Crime: FBI UCR (Kaplan); Permits: Census BPS.
- April 1 alignment between rate panel and LODES reference date.

Summary Statistics

Variable	Mean	Median	SD	Max
Panel A: Pennsylvania (2002–2019)				
share_live_in	0.149	0.127	0.097	0.700
Resident EIT rate (%)	0.542	0.500	0.199	4.331
Total jobs	3,624	1,224		
Resident workers	674	154		
Panel B: Ohio (2002–2019)				
share_live_in	0.119	0.099	0.085	
Municipal tax rate (%)	1.222	1.500	0.821	3.000
PA muni × year	26,650 (1,479 munis)			
OH muni × year	7,740 (430 munis)			

Notes: Sample restricted to munis with > 50 total jobs.

Treatment Geography: PA Rate Changes



Red = increased; blue = decreased; purple = both; grey = unchanged.

Three Channels of Adjustment

When muni A raises its rate, workers in A can:

1. Move residence only (keep job in A). Under credit system: liability still τ_A. Zero tax savings, but reduces `share_live_in`.
2. Move residence and job to lower-rate B. New liability τ_B. Canonical Tiebout margin.
3. New hires choose to live outside A. Reduces share even without existing-resident moves.

Predictions we'll test: (1) enforcement matters $\Rightarrow$ post-Act 32 effect; (2) destination rate irrelevant for commuters; (3) complete relocators pick lower-rate destinations; (4) SD rate opposite sign; (5) boundary gradient.

Q1: Does tax adoption cause sorting?

Model (Sun–Abraham, pooled OH+PA):

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \cdot D_{it}^{\ell} + \varepsilon_{it}$$

- Y_{it} = `share_live_in`; D_{it}^{ℓ} = event time relative to first adoption.
- Treated: 8 adopters (1 OH, 7 PA); Control: 109 never-taxed munis.
- State × year FE; min 10 total jobs.

Cleanest extensive-margin test — but tiny treated cell.

Q1 Answer: Directionally consistent, underpowered

	Pooled (OH+PA)
Sun–Abraham ATT	−0.0036
	(0.0066)
Treated / Control	8 / 109

- Sign is right but we cannot reject zero.
- Only 8 adopters with enough LODES employment ⇒ insufficient power for the extensive margin.
- Move to rate changes among already-taxed cities.

Q2: Do rate increases cause sorting?

Same Sun–Abraham spec, plus DCDH continuous treatment:

$$Y_{it} = \alpha_i + \gamma_t + \sum_{\ell \neq -1} \mu_\ell \cdot D_{it}^{\ell} + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

- Treated: first rate increase; two control groups: (i) no-tax, (ii) stable-rate.
- **?** for continuous rate + placebo pre-trends.
- PA fiscal controls $\mathbf{X}$: police, streets, debt, state aid.

Q2 Answer: −0.5 to −0.8 share-points per 1 pp rate hike

	Inc vs No-Tax		Inc vs Stable	
	OH	PA	OH	PA
SA ATT	-0.0074^{***}	-0.0091^{*}	−0.0001	-0.0053^{*}
	(0.0025)	(0.0046)	(0.0021)	(0.0029)
DCDH placebo p	0.406	—	0.147	0.904

- Rate increases reduce `share_live_in` by −0.005 to −0.008 per 1 pp of rate increase, robust across estimators + states.
- OH: no-tax townships are better controls than stable-rate cities (latter share confounding urban traits).
- PA DCDH: clean pre-trends ($p = 0.90$); effect -0.011^{***} at $\ell = 8$.

Mechanism: Is it enforcement? (Act 32)

Structural break on PA rate effect:

$$Y_{it} = \alpha_i + \gamma_t + \beta^{\text{pre}} \tau_{it} \cdot \mathbf{1}[t < 2012] + \beta^{\text{post}} \tau_{it} \cdot \mathbf{1}[t \geq 2012] + \varepsilon_{it}$$

- Prediction 1: response requires a binding tax.
- Pre-2012: fragmented collection, weak nonresident enforcement.
- Post-2012: centralized TCDs + employer withholding at $\max(\tau^{\text{res}}, \tau^{\text{work}})$.

Act 32 Answer: Sorting only once the tax is binding

	Split Periods	Interaction
$\tau \times \mathbf{1}[t < 2012]$	-0.0021 (0.0044)	
$\tau \times \mathbf{1}[t \geq 2012]$	-0.0179^{***} (0.0051)	
τ		-0.0021 (0.0044)
$\tau \times$ Post-2012		-0.0158^{***} (0.0034)
N	27,977	27,977

- Precise zero before; after 2012, a 1 pp rate hike cuts `share_live_in` by 1.8 share-points — same statutory rate.
- Difference significant at 1%. Workers respond to taxes they actually pay, not paper rates.
- Confirms fiscally: same rate generates 5.1% more EIT revenue post-2012.

Q3: Are rate changes symmetric?

Split cumulative dose:

$$Y_{it} = \alpha_i + \gamma_t + \beta^+ R_{it}^+ + \beta^- R_{it}^- + \mathbf{X}_{it}'\boldsymbol{\phi} + \varepsilon_{it}$$

- R_{it}^+, R_{it}^-: cumulative increases / decreases from baseline.
- If $|\beta^+| > |\beta^-| \Rightarrow$ residential choices are sticky (fixed costs of moving).
- Also estimate joint inc/dec event indicators.

Q3 Answer: Sharply asymmetric

	PA Joint	PA Split Dose
inc × post	−0.0070** (0.0034)	
dec × post	−0.0000 (0.0058)	
R^+ (cum. inc)		−0.0177** (0.0069)
R^- (cum. dec)		+0.0012 (0.0040)

- Increases: −1.8 share-points per 1 pp rate hike; decreases: null.
- Holds across both states and all estimators.
- Interpretation: fixed moving costs ⇒ once pushed out, residents do not return with a cut.
- Policy: a city that raises then lowers its rate will not recover the residents it lost.

Q4: Leave entirely, or just change where they sleep?

Resident worker panel (complete relocation):

$$\ln(\text{ResWorkers}_{Bt}) = \alpha_B + \gamma_t + \beta_1 \tau_{Bt} + \beta_2 \bar{\tau}_{Bt}^{\text{nbr}} + \varepsilon_{Bt}$$

Commuter pair model (Channel 1 tax optimization?):

$$\ln(\text{Flow}_{ABt}) = \alpha_{AB} + \gamma_t + \beta_1 \tau_{At}^{w} + \beta_2 \tau_{Bt}^{h} + \varepsilon_{ABt}$$

- $\beta_2 > 0$ in first eq. $\Rightarrow$ neighbors absorb displaced workers (Channel 2).
- $\beta_2 \approx 0$ in second eq. $\Rightarrow$ credit system works as designed (Prediction 2).

Q4 Answer: Cross-boundary relocation, not tax arbitrage

	Resident Workers	Commuter Flow
Own EIT τ_B / τ_A^w	-0.140^{**} (0.063)	-0.043 (0.031)
Neighbor avg rate	$+1.069^{*}$ (0.593)	—
Home rate τ_B^h	—	-0.001 (0.006)
Own EIT × Post-2012	-0.210^{***} (0.062)	-0.059^{*} (0.034)
Own EIT × Pre-2012	-0.042 (0.044)	-0.020 (0.023)

- Higher own rate $\Rightarrow -14\%$ log resident workers; neighbors gain when you raise rates.
- Home rate irrelevant for commuters (Prediction 2 confirmed).
- Effects $\sim 5\times$ larger post-Act 32 across both panels.

Q5: Do school district rates have the opposite effect?

Overlapping-tax decomposition:

$$Y_{it} = \alpha_i + \gamma_t + \beta^m \tau_{it}^m + \beta^{sd} \tau_{it}^{sd} + \varepsilon_{it}$$

- Muni rate raised for fiscal needs; SD rate (Act 1 of 2006) for property tax relief / school spending.
- Muni–SD correlation only $r = -0.12 \Rightarrow$ independent variation.
- Prediction 4: $\beta^m < 0$ (repulsion), $\beta^{sd} > 0$ (amenity).
- Also estimate on commuter pairs with rates at both ends.

Q5 Answer: Opposite signs — SD attracts residents

	log(Resident)	log(Nonresident)	log(Total Jobs)
Muni EIT τ^m	-0.129^{**}	-0.073^{*}	-0.091^{**}
	(0.060)	(0.042)	(0.042)
SD EIT τ^{sd}	$+0.161^{***}$	$+0.080$	$+0.080$
	(0.049)	(0.055)	(0.050)
Total EIT	$+0.001$ (0.034) — zero net effect		

- SD operates only on residents; not commuters / total jobs ⇒ residential amenity channel.
- Total EIT (muni+SD) nets to zero ⇒ explains prior null findings that aggregate the two rates.
- Commuter pairs: home SD rate $+0.088^{***}$ (people live where schools are funded).

Q6: Is sorting concentrated at municipal boundaries?

Tract-level within-city model:

$$S_{jt}^{\text{within}} = \alpha_j + \gamma_t + \beta_1 \tau_{it} + \beta_2 \tau_{it} \times \mathbf{1}[\text{boundary}_j] + \varepsilon_{jt}$$

- Tract j in muni i; boundary = centroid within 1 km of nearest boundary.
- S_{jt}^{within} = share of workers at tract j whose home is also in i.
- Prediction 5: $\beta_2 < 0$ if cross-boundary arbitrage dominates.
- Also estimate a distance-to-boundary gradient (0–0.5, 0.5–1, 1–2, 2–5, 5+ km).

Q6 Answer: No boundary gradient — city-wide response

	Within-city share	log(Total workers)
Muni EIT	$+0.0006^{**}$ (0.0003)	$+0.0217^{***}$ (0.0027)
Muni EIT × Boundary	$+0.0008$ (0.0013)	-0.0449^{*} (0.0266)

- Within-city reshuffling: essentially zero ⇒ people are crossing muni boundaries, not just neighborhoods.
- Boundary × rate on share: null. Boundary tracts respond like interior tracts.
- Boundary tracts do lose total workers marginally ($-4.5\%^{*}$), hinting at employment leaking across boundaries.
- Consistent with whole-city relocation, not short-distance tax arbitrage.

Q7: Where does the incremental revenue go?

Expenditure decomposition (AFR):

$$\ln(\text{Exp}_{it}^{k}) = \alpha_i + \gamma_t + \beta^k \tau_{it} + \varepsilon_{it}$$

$k \in \{\text{police, fire, streets, admin, debt, sewer, rec, comm dev}\}$.

Does visibility of spending moderate sorting?

$$Y_{it} = \alpha_i + \gamma_t + \beta_1 \tau_{it} + \beta_2 \tau_{it} \times S_i^{\text{visible}} + \varepsilon_{it}$$

S_i^{visible} = pre-treatment share on police + fire + streets.

Q7 Answer: Revenue funds invisible community dev

	Police	Streets	Comm Dev	Fire
τ^m	0.051	0.081*	0.915***	0.080
	(0.046)	(0.042)	(0.303)	(0.051)

- Only community development (planning, econ dev, housing) responds strongly — long-horizon, hard to see.
- Police / fire: positive but imprecise. Residents do not directly see more patrols.
- Visibility interaction +0.024 (SE 0.047): suggestive, not significant — 171 treated munis limits power.
- SD side (PDE): rate increases reallocate (facilities -0.73^{***}, admin -0.03^{**}), not expand.

Fiscal Impact: Rates "Work"

	log(EIT)	log(Tot. Tax)	log(Tot. Rev)	log(Police)	log(Pop)	log(MV)
τ^{res}	0.671^{***}	0.296^{***}	0.064	0.053	-0.009	0.007
	(0.211)	(0.095)	(0.043)	(0.046)	(0.014)	(0.022)

- EIT revenue $+67\%^{***}$, total tax $+30\%^{***}$ per 1 pp of rate increase.
- Population, property values, total revenue: null.
- Behavioral response too small to erode the base — rate increases are on the right side of the revenue hill.

Robustness

- Pre-trends clean across 11 observables: revenue, EIT rev, RE tax, streets, pop, debt, jobs, crime, permits. Exceptions (police, market value) addressed with controls / coverage notes.
- Fiscal controls (log police, streets, debt, state aid) leave point estimates and SEs essentially unchanged.
- Philadelphia exclusion: results robust to dropping Philly.
- Flip-floppers: 37% of PA treated munis flip-flop; excluding them strengthens results.
- Placebo: Federal workers (JT04) show no response (federal wages not subject to local withholding same way).
- ACS / IRS SOI: county-level migration null — LODES detects what coarser data cannot.

Putting it Together (1/2): Who moves?

Setup. A post-Act 32 PA city: 1,000 total workers; 150 live-and-work there (share_live_in = 15%); EIT rises 0.5% → 1.5%.

Q2/Q3. Over 7–8 years, share falls to ∼13.2% ⇒ ∼18 of 150 locally-resident workers gone. If the city later cuts the rate back, none of them return.

Q4. Those 18 leave the city entirely (not a neighborhood move); nearby lower-rate munis within 80 km absorb them.

Q5. If the overlapping school district had raised its rate instead, residents would have moved in, not out.

Q6. The loss is spread across the city, not concentrated at municipal boundaries — no "edge effect."

Putting it Together (2/2): Was it worth it?

Same city: 1,000 workers, rate 0.5% → 1.5%, ∼18 residents lost.

Fiscal. EIT revenue roughly doubles (+67 log points); total tax +30%. Property values and population: unchanged.

Q7. Most new dollars flow to community development (planning, econ dev, housing) — not to police patrols residents see.

nforcement. None of this happens pre-Act 32: same statutory rate, precise zero behavioral response. Collection capacity is the binding constraint.

Bottom line. The city trades ∼18 residents for a near doubling of EIT revenue. Rate increases work — when enforced.

Summary

1. Local income tax rate increases cause residential sorting: `share_live_in` falls by 0.5–0.8 share-points per 1 pp rate hike, building over 7–8 years.
2. Effect is enforcement-contingent: zero pre-Act 32, large post-2012.
3. Sorting is asymmetric (sticky residences) and city-wide (not boundary arbitrage).
4. Muni and SD rates have opposite signs — SD acts as an amenity.
5. Rate increases are fiscally productive: base erosion too small to offset revenue gains.

Next Steps

- Link Act 1 SD rate changes to actual property tax millage movements (PDE data 2015+ restrictive).
- Extend Q7 with richer, pre-treatment expenditure composition (more power on visibility channel).
- Apply the LODES-based design to other local instruments (property tax millage, local sales taxes, fees).
- Welfare: combine behavioral elasticity with revenue elasticity to bound efficiency cost of local income tax.

Thank You

Questions?

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github.com/MatthewHockert/local-income-tax-migration